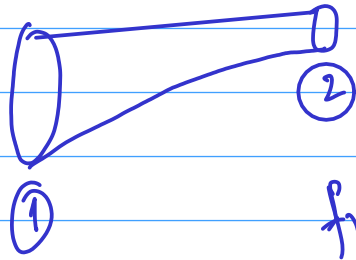


Fluid Flow



Equivalent Diameter

$$d_h = \frac{4A}{P}$$

Cross-section Area
↓
Perimeter

frictionless flow

$$\left(\frac{P}{\rho}\right)_1 + \frac{V_1^2}{2} + gz_1 = \left(\frac{P}{\rho}\right)_2 + \frac{V_2^2}{2} + gz_2 \quad \left[\frac{J}{kg}\right]$$

$$\left(\frac{P}{\rho} + \frac{V^2}{2} + gz\right)_1 - \left(\frac{P}{\rho} + \frac{V^2}{2} + gz\right)_2 = \text{loss of energy}$$

with friction

$$\left(\frac{P}{\rho}\right)_1 + \frac{V_1^2}{2} + gz_1 = \left(\frac{P}{\rho}\right)_2 + \frac{V_2^2}{2} + gz_2 + \underbrace{\left[f \frac{L V^2}{d^5} + K_L \frac{V^2}{2} \right]}_{\text{losses}}$$

major loss minor loss

pipe roughness $\left(\frac{\epsilon}{d}\right)$ value
↓
dia.

$$f = f_{\text{un}}\left(Re_d, \frac{\epsilon}{d}\right)$$

major loss →

density pipe dia
↓ ↓
ρ d

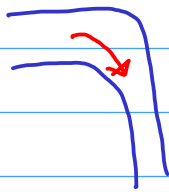
friction factor

$$Re_d = \frac{\rho V d}{\mu} = \frac{V d}{\nu}, \quad \nu = \frac{\mu}{\rho}$$

absolute viscosity

kinematic viscosity

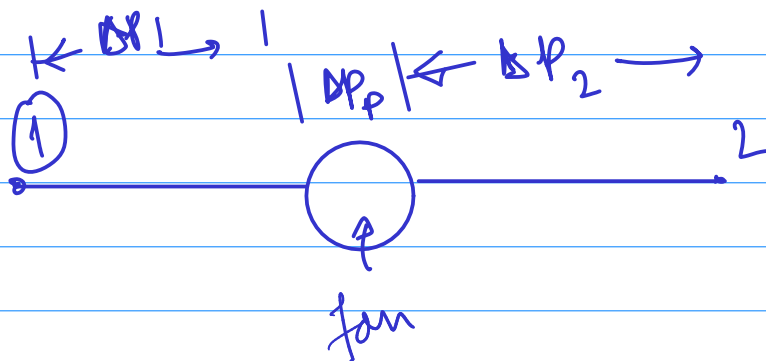
minor loss $\rightarrow$



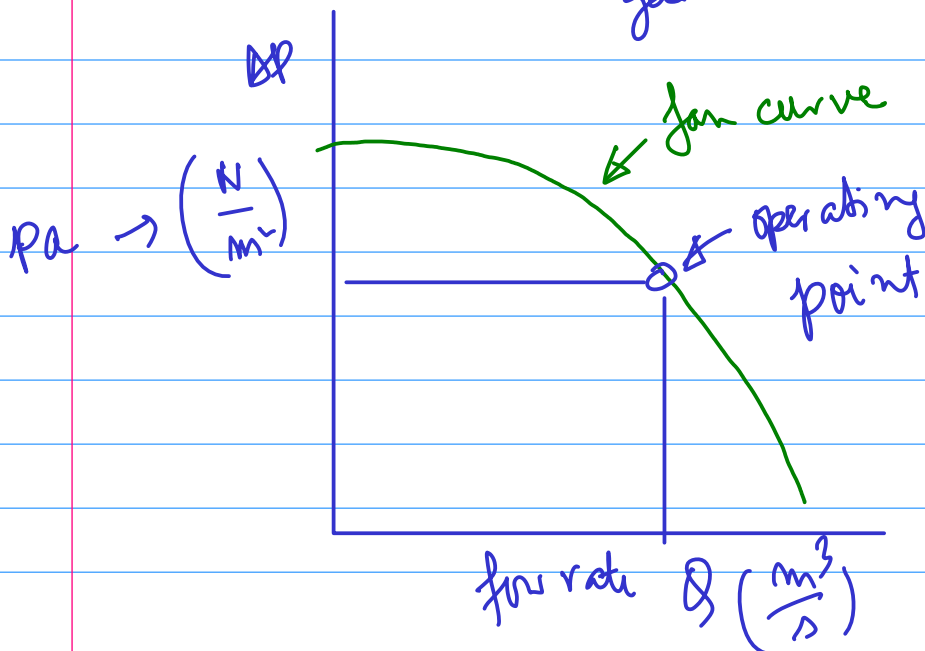
from handbook



$$K_L = \text{fun}(\text{geometry, vel})$$

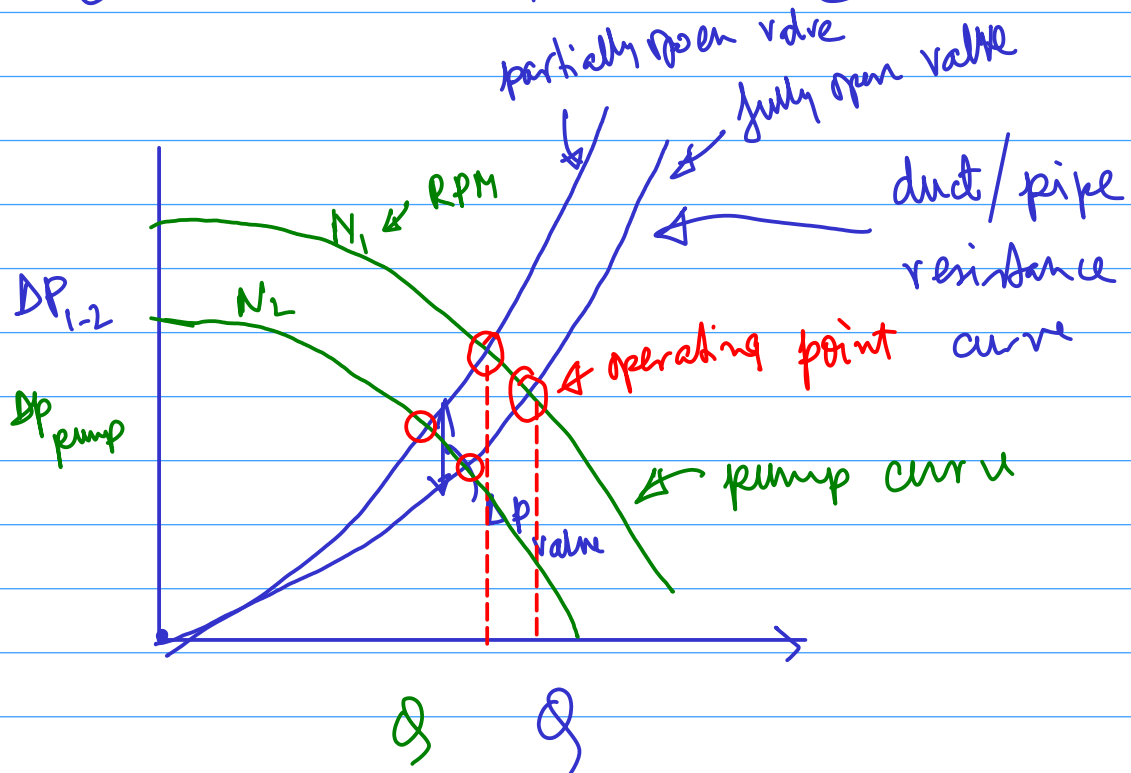
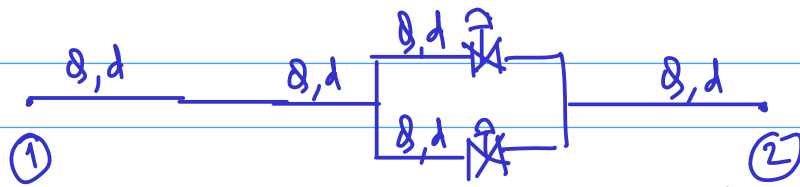


$$\left(\frac{p}{\rho}\right)_1 + \frac{V_1^2}{2} + gz_1 + \underbrace{\frac{1}{\rho} \Delta P_{\text{pump or fan}}}_{\text{gain}} = \left(\frac{p}{\rho}\right)_2 + \frac{V_2^2}{2} + gz_2 + \underbrace{\left(\frac{f}{d} L \frac{V^2}{2} + K_L \frac{V^2}{2}\right)}_{\text{loss}}$$



fan curve

$$Q = \frac{\pi}{4} d^2 \times V_{\text{avg}}$$



1st law of thermodynamics (SSSF)

$$\left(h + \frac{V^2}{2} + gz\right)_1 = \left(h + \frac{V^2}{2} + gz\right)_2 - \dot{W}_{in}$$

$$\underset{\substack{\uparrow \\ \text{enthalpy}}}{h} = \underset{\substack{\uparrow \\ \text{internal energy}}}{u} + \underset{\substack{\uparrow \\ \text{sp. vol.}}}{p v} = \underset{\substack{\uparrow \\ \text{internal energy}}}{u} + \frac{p}{\rho}$$

$$\left(\frac{p}{\rho} + \frac{V^2}{2} + gz\right)_1 + u_1 + \frac{\Delta p_{fan}}{\rho} = \left(\frac{p}{\rho} + \frac{V^2}{2} + gz\right)_2 + u_2$$

$$u_1 - u_2 = \underset{\substack{\uparrow \\ \text{sp. heat at constant volume}}}{C_v}(T_1 - T_2)$$

$$\left(\frac{p}{\rho} + \frac{V^2}{2} + gz \right)_1 + u_1 + \frac{\Delta p_{\text{fam}}}{\rho} = \left(\frac{p}{\rho} + \frac{V^2}{2} + gz \right)_2 + u_2$$

$$u_1 - u_2 = c_v (T_1 - T_2)$$

↑
sp. heat at constant volume

Accounting for major/minor losses

heat trans.
↓

$$\left(\frac{p}{\rho} + \frac{V^2}{2} + gz \right)_1 + \frac{\Delta p_{\text{fam}}}{\rho} = \left(\frac{p}{\rho} + \frac{V^2}{2} + gz \right)_2 + c_v (T_2 - T_1)$$

mechanical
energy.

$$+ f \left(\frac{L}{d} \right) \left(\frac{V^2}{2} \right) + K_L \left(\frac{V^2}{2} \right)$$

major
loss

minor loss
due to
fittings

Finally obtained from
1st law of thermodynamics
for duct flow.

friction factor f
loss coefficient K_L

→ from charts/tables